

Review Notes for Multi Stochastic Trends Systematic Factors Extension

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1 Overall Assessment

The model setup, propositions, and appendix proofs are largely correct and internally consistent. The main structure—linking the pass-through matrix Λ_r to the cointegration rank and crisis dynamics—is mathematically sound. However, there are several notational inconsistencies and a few conceptual issues that should be clarified. This note documents them carefully.

2) Pricing kernel / global factors (SDF)

What's there: $g_t = (g_{1,t}, \dots, g_{q_r,t})^\top$ are independent $I(1)$ global factors with regime-dependent drifts/vols; the SDF shock is

$$\xi_{t+1} = \lambda_r^\top g_{t+1} + v_{t+1}$$

and the total SDF innovation variance is

$$\varsigma_r^2 = \lambda_r^\top \Sigma_{g,r} \lambda_r + \sigma_v^2.$$

The factor random walks are stated explicitly:

$$g_{k,t+1} = g_{k,t} + \mu_{k,r} + \sigma_{g_k,r} \varepsilon_{k,t+1}.$$

Verdict: Mathematically consistent. The log-normal SDF/Jensen correction is later derived cleanly (Appendix eq. (21)):

$$\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \xi_{t+1}, \quad E_t[M_{t+1}] = e^{-r_t}.$$

Minor fixes: You assume independent factor innovations via diagonal $\Sigma_{g,r}$. This is fine for exposition, but independence is not required; you could allow a full positive-definite covariance and keep all propositions intact (you already use the general form in the Prop. 2 proof).

3) Payoffs, no-arbitrage, and FX restriction

What's there: For a pure FX claim, $P_{i,t} = 1/S_{i,t}$ and the key restriction

$$1 = E_t[M_{t+1} \exp(X_{i,t} - X_{i,t+1})],$$

with $X_{i,t} \equiv \log S_{i,t}$.

Verdict: Correct and standard.

4) Regime-dependent pass-through (“autonomy”) and Λ_r

What's there: Loadings are $\lambda_{i,r} = (\mathbf{1} - \psi_{i,r}) \odot \bar{\lambda}_i$ with $\psi_{i,r} \in [0, 1]^{q_r}$; you stack them into the $N \times q_r$ pass-through matrix Λ_r . Assumptions A1–A4 govern vol spikes and autonomy compression in crisis and partial recovery post-crisis.

Verdict: Internally consistent and ties directly to later rank/correlation results.

Two conceptual cautions & edits:

- Full autonomy ($\psi_{i,r} = 1$ for all k) forces $\lambda_{i,r} = 0$, making $X_{i,t}$ stationary. To preserve $I(1)$, assume: for every i, r , at least one component of $\psi_{i,r} < 1$ (equivalently $\lambda_{i,r} \neq 0$).
- Spell out that $\bar{\lambda}_i \neq 0$ (already noted) and that ψ rescales currency loadings, not the SDF itself—your appendix derivations already distinguish λ_r (SDF) and $\lambda_{i,r}$ (asset).

6) Exchange-rate system / rank mapping

What's there: You collect into

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

and state the rank mapping: cointegration rank = $N - \text{rank}(\Lambda_r)$; special case $q_r = 1$ gives $N - 1$ if $\Lambda_r \neq 0$.

Verdict: Correct and clean. Also consistent with the appendix proofs.

Key propositions vs. setup (consistency check)

- **Prop 1 (rank/cointegration):** Matches $N - \text{rank}(\Lambda_r)$.
- **Prop 2 (alignment in crisis):** Proof formalizes long-horizon covariance scaling with $\lambda_i^\top (\Sigma_{g,r} \Sigma_{g,r}^\top) \lambda_j$. Correct, but limiting correlation $\rightarrow \pm 1$ only if loadings are collinear; otherwise correlation increases but need not approach 1.
- **Prop 3 (post-crisis decoupling for EMs):** Follows from $\lambda_{i,r} = (1 - \psi_{i,r}) \bar{\lambda}_i$ and A1–A3.
- **Prop 4 (ECM half-life):** Standard ECM result; consistent.

2 Verification of Propositions

This section reviews the four propositions stated in the paper. For each, I restate the proposition, provide or draft a brief independent proof from the model setup, and then compare it with the corresponding proof in the appendix. Any issues, ambiguities, or necessary corrections are highlighted.

Proposition 1 (Rank–Cointegration Link)

Statement. Given regime r and assumptions A1–A4, the cointegration rank of X_t is

$$\text{Cointegration rank} = N - \text{rank}(\Lambda_r).$$

Independent proof. From the model setup,

$$X_t = \mu_r + \Lambda_r g_t + \Gamma z_t + u_t,$$

with $g_t \sim I(1)$, and z_t, u_t stationary. For any $b \in \mathbb{R}^N$,

$$b' X_t = b' \mu_r + (b' \Lambda_r) g_t + b' \Gamma z_t + b' u_t.$$

The last two terms are stationary. Thus $b'X_t$ is stationary iff $b'\Lambda_r = 0$. Hence the cointegration space is $\{b : b'\Lambda_r = 0\}$ of dimension $N - \text{rank}(\Lambda_r)$.

Comparison with appendix. The appendix gives essentially the same argument. The result is correct. **Note:** the paper does not mention the edge case $\Lambda_r = 0$; in that case cointegration rank is N , not $N - 1$.

Proposition 2 (Crisis Co-Movement Amplification)

Statement. If $\text{rank}(\Lambda_r)$ remains unchanged in crisis, but factor volatilities rise and autonomy compresses (loadings align), then long-run correlations increase.

Independent proof. From the difference form,

$$\Delta X_{i,t} = \lambda'_{i,r} \Delta g_t + \Delta(\Gamma'_i z_t + u_{i,t}).$$

Summing over H horizons,

$$Y_{i,H} = \sum_{h=1}^H \Delta X_{i,t+h} = \lambda'_{i,r} (g_{t+H} - g_t) + \text{stationary}.$$

Thus, as $H \rightarrow \infty$,

$$\frac{1}{H} \text{Cov}(Y_{i,H}, Y_{j,H}) \rightarrow \lambda'_{i,r} \Sigma_{g,r} \Sigma'_{g,r} \lambda_{j,r},$$

$$\frac{1}{H} \text{Var}(Y_{i,H}) \rightarrow \lambda'_{i,r} \Sigma_{g,r} \Sigma'_{g,r} \lambda_{i,r}.$$

Hence the long-run correlation is

$$\rho_{ij} = \frac{\lambda'_{i,r} \Sigma_{g,r} \Sigma'_{g,r} \lambda_{j,r}}{\sqrt{(\lambda'_{i,r} \Sigma_{g,r} \Sigma'_{g,r} \lambda_{i,r})(\lambda'_{j,r} \Sigma_{g,r} \Sigma'_{g,r} \lambda_{j,r})}}.$$

Rising volatility increases the dominance of common shocks over stationary terms. If autonomy compresses, the loading vectors $\lambda_{i,r}, \lambda_{j,r}$ become more collinear, and ρ_{ij} rises. It tends toward ± 1 only if the vectors are proportional.

Comparison with appendix. The appendix proof derives the same limiting form but states that correlation approaches ± 1 without qualification. *Correction:* this is strictly true only if loadings are collinear. In general, correlation increases but need not approach ± 1 .

Proposition 3 (Rank Change in Crisis)

Statement. If new persistent global factors emerge or autonomy shifts, then $\text{rank}(\Lambda_{\text{crisis}}) > \text{rank}(\Lambda_{\text{pre}})$, reducing the number of cointegrating relations. If $\text{rank}(\Lambda_{\text{crisis}}) = N$, cointegration disappears entirely.

Independent proof. Since cointegration rank is $N - \text{rank}(\Lambda_r)$, adding new independent stochastic trends or altering loadings to span more dimensions increases $\text{rank}(\Lambda_r)$, reducing cointegration rank. If $\text{rank}(\Lambda_r) = N$, then no nontrivial b satisfies $b'\Lambda_r = 0$, hence no cointegration remains.

Comparison with appendix. The appendix proof is identical in logic. This result is fully correct.

Proposition 4 (Post-Crisis Divergence)

Statement. If $\text{rank}(\Lambda_{\text{post}}) < \text{rank}(\Lambda_{\text{crisis}})$ but dispersion in autonomy parameters widens, then long-run correlations weaken.

Independent proof. A reduction in rank means fewer stochastic drivers, restoring some cointegration. But if $\psi_{i,r}$ diverge, then the cosine similarity between loading vectors $\lambda_{i,r}$ falls, so cross-currency correlations weaken even though some cointegration relations reappear.

Comparison with appendix. The appendix proof for the $q_r = 1$ case shows exactly this effect: larger ψ post-crisis reduces loadings, increasing dispersion and weakening correlations. This is consistent with the independent proof.

Summary of Proof

1. Proposition 1 is correct, but the edge case $\Lambda_r = 0$ (all currencies autonomous) is not mentioned. In that case the cointegration rank is N .
2. Proposition 2 is correct in the single-factor case, but the general statement is overstated. In multi-factor cases correlations increase, but they approach ± 1 only if loadings are collinear.
3. Propositions 3 and 4 are correct and consistent with the model setup.

4. Appendix proofs are largely correct, but Proposition 2's proof omits the collinearity qualification, and Proposition 1 omits the $\Lambda_r = 0$ edge case.

Short list of actionable edits

1. Explicitly require $\lambda_{i,r} \neq 0$ (i.e. at least one $\psi_{i,r} < 1$) to preserve $I(1)$ exchange-rate dynamics. Add an $I(1)$ safeguard: for each i, r , require at least one $\psi_{i,r} < 1$ so $\lambda_{i,r} \neq 0$.
2. Notation polish: state $\mu_r = (\mu_{1,r}, \dots, \mu_{q_r,r})^\top$, you indicated in eq 12, but needed to be explicit to claim it's drift vector to align well with eq 3.; allow $\Sigma_{g,r}$ to be full and general, not necessarily diagonal.
3. Distinguish λ_r (SDF parameter) and $\lambda_{i,r}$ (currency loadings) more carefully.
4. Prop. 2 wording: "increases toward ± 1 when loadings are collinear; otherwise increases but need not approach 1."
5. Explicitly mention the edge case $\Lambda_r = 0$ (all currencies autonomous, no cointegration)